

(A3) There exists a constant $c_L > 0$ such that for all $t \in [0, T]$, $x'_0, x_0 \in \mathbb{R}^{d_0}$, $\alpha'_0, \alpha_0 \in A_0$, $x', x \in \mathbb{R}^d$, $\alpha', \alpha \in A$, $\mu', \mu \in \mathcal{P}_2(\mathbb{R}^d)$, we have:

$$\begin{aligned} & |(f_0, g_0)(t, x'_0, \alpha'_0, \mu') - (f_0, g_0)(t, x_0, \alpha_0, \mu)| \\ & \leq c_L (1 + |x'_0| + |x_0| + |\alpha'_0| + |\alpha_0| + M_2(\mu') + M_2(\mu)) \\ & \quad \times (|x'_0 - x_0| + |\alpha'_0 - \alpha_0| + W_2(\mu', \mu)). \end{aligned}$$

and

$$\begin{aligned} & |(f, g)(t, x_0, x', \alpha_0, \alpha', \mu') - (f, g)(t, x_0, x, \alpha_0, \alpha, \mu)| \\ & \leq c_L (1 + |x'_0| + |x_0| + |x'| + |x| + |\alpha'_0| + |\alpha_0| + |\alpha'| + |\alpha| + M_2(\mu') + M_2(\mu)) \\ & \quad \times (|x'_0 - x_0| + |x' - x| + |\alpha'_0 - \alpha_0| + |\alpha' - \alpha| + W_2(\mu', \mu)). \end{aligned}$$

(A4) The maps $(x_0, \alpha_0) \mapsto b_0, b, f_0$ are differentiable, and the maps $(x_0, \alpha_0, \mu) \mapsto \partial_{x_0}(b_0, b, f_0)$ and $(x_0, \alpha_0, \mu) \mapsto \partial_{\alpha_0}(b_0, b, f_0)$ are continuous. The maps $(x, \alpha) \mapsto b, \sigma, f$ are differentiable, and the maps $(x, \alpha, \mu) \mapsto \partial_x(b, \sigma, f)$ and $(x, \alpha, \mu) \mapsto \partial_\alpha(b, f)$ are continuous.

The maps $\mu \mapsto b_0, b, f_0$ are L differentiable, and the maps $(x_0, X, \alpha_0) \mapsto \partial_\mu(b_0, f_0)(t, x_0, \alpha_0, \mathcal{L}(X))(X)$ are continuous.

The map $x_0 \mapsto g_0$ is differentiable, and the map $(x_0, \mu) \mapsto \partial_{x_0}g_0$ is continuous. The map $\mu \mapsto g_0$ is L differentiable, and the map $(x_0, X) \mapsto \partial_\mu g_0(x_0, \mathcal{L}(X))(X)$ is continuous. The map $x \mapsto g$ is differentiable, and the mapping $(x_0, x, \mu) \mapsto \partial_x g$ is continuous.

2.2. Finite player setting. The [MmMFG strategy](#) is meant to serve as a mean field description of the following finite player framework, which is described through a stochastic differential game. Suppose we have $N + 1$ players, where the 0-th player corresponds to the major player, and players $1, \dots, N$ are the minor players. Let $\mathbf{B} = (B^1, \dots, B^N)$ denote a Nd -dimensional Brownian motion. Denote by $\mathbb{A}_i$ the space of admissible strategies for the i -th minor player, namely strategies $\alpha^i : [0, T] \times \Omega \rightarrow A$ which are progressively measurable w.r.t. the augmented filtration generated by B^i , and $\mathbb{E}[\int_0^T |\alpha_t^i|^2 dt] < \infty$. The major player's admissible strategies remain elements of $\mathbb{A}_0$. Suppose the players possess the following state processes:

$$\begin{aligned} dX_{0t} &= b_0(t, X_{0t}, \alpha_{0t}, \bar{\mu}_{\mathbf{X}_t})dt, \\ dX_t^i &= b(t, X_{0t}, X_t^i, \alpha_{0t}, \alpha_t^i, \bar{\mu}_{\mathbf{X}_t^{-i}})dt + \sigma(t, X_t^i)dB_t^i, \quad X_0^i = x_0^i, \quad i = 1, \dots, N, \end{aligned}$$

where X_0 refers to the state process of the major player, and X^i refers to the state process of the i -th minor player. Moreover, we write $\mathbf{X} := (X^1, \dots, X^N)$ and $\boldsymbol{\alpha} := (\alpha^1, \dots, \alpha^N)$.

The utility function for the major player is given by:

$$J_0(\alpha_0, \boldsymbol{\alpha}) = \mathbb{E} \left[\int_0^T f_0(t, X_{0t}, \alpha_{0t}, \bar{\mu}_{\mathbf{X}_t})dt + g_0(X_{0T}, \bar{\mu}_{\mathbf{X}_T}) \right],$$

and for the i -th minor player:

$$J^i(\alpha_0, \alpha^i, \boldsymbol{\alpha}^{-i}) = \mathbb{E} \left[\int_0^T f(t, X_{0t}, X_t^i, \alpha_{0t}, \alpha_t^i, \bar{\mu}_{\mathbf{X}_t^{-i}})dt + g(X_{0T}, X_T^i, \bar{\mu}_{\mathbf{X}_T^{-i}}) \right], \quad i = 1, \dots, N.$$

The objective of the above $N + 1$ player game is for the major player to minimise $\alpha_0 \mapsto J_0(\alpha, \boldsymbol{\alpha})$ and the minor players to minimise $\alpha^i \mapsto J^i(\alpha_0, \alpha^i, \boldsymbol{\alpha}^{-i})$ in the sense of Nash equilibrium. However, finding true Nash equilibria to this stochastic differential game is difficult. Put simply, this